VOEDING 12 V

J1 — 12V IN (GND, VIN_RAW)

Q1 — AO4407A (GS_G, GND)

D2 — BZT52C15 (12V, GS_G)

D1 — SMBJ15CA (12V, GND)

C4 — 330µ/25V

C5 — 330µ/25V

C6 — 100n/25V

R5 — 100k (GS_G, GND)

R6 — 100k (VIN_RAW, GS_G)

C7 — 100n (VIN_RAW, GND)

BUCK 12 V –> 5 V 5 A

U1 — TPS554608 (12V, EN, COMP, RT/CLK, GND, BOOT, SW, FB)

C8 — 100n/25V (12V, GND)

C9 — 100n/25V (12V, GND)

C10 — 100n (12V, GND)

R7 — 47k (BUCK_EN, 12V)

R8 — 100k (BUCK_RT, GND)

C11 — 220p (BUCK_RT, SW)

C12 — 10n (SW, BUCK_FB)

R9 — 68k (BUCK_FB, BUCK_COMP)

D4 — SS54 (SW, GND)

C1 — 47µ/50V (SW, GND)

C2 — 47µ/50V (SW, GND)

C3 — 47µ/50V (SW, GND)

L1 — 6µ8 (SW, BUCK_A)

R1 — 100k (BUCK_A, GND)

R2 — 22k (BUCK_A, BUCK_COMP)

R3 — 10k (BUCK_COMP, GND)

C13 — 2n2 (BUCK_A, BUCK_COMP)

C14 — 100n (BUCK_COMP, GND)

D5 — LED 5V (GND, LED5_A)

R4 — 1k (LED5_A, BUCK_A)

J13 — 5V PI (GND, 5V)

3,3 V (ESP32 + IMU)

D6 — SS14 (VS_ESP, VS)

D7 — SS14 (VS_ESP, VBUS_ESP)

U2 — AMS1117-3.3 (VS_ESP, 3V3, GND)

C15 — 22µ (3V3, GND)

C16 — 100n (3V3, GND)

C17 — 10µ (3V3, GND)

ESP32-S3 (SERVO-BUS + IMU + USB NAAR PI)

U3 — ESP32-S3-WROOM-1-N16R8 (3V3, GND, EN, IO0, IO1, IO2, IO3, IO4, IO5, IO6, IO7, IO8, IO9, IO10, IO11, IO12, IO13, IO14, IO15, IO16, IO17, IO18, IO19, IO20, IO21, IO35, IO36, IO37, IO38, IO39, IO40, IO41, IO42, IO45, IO46, IO47, IO48, TXD0, RXD0, USB_D-, USB_D+)

D8 — LED ESP (GND, LED_ESP_A)

R10 — 10k (3V3, ESP_EN)

C18 — 100n (ESP_EN, GND)

R11 — 10k (3V3, IO0)

C19 — 100n (IO0, GND)

R12 — 330 (LED_ESP_A, IO21)

R13 — 10k (IMU_SDA, 3V3)

R14 — 10k (IMU_SCL, 3V3)

R15 — 10k (IMU_INT, 3V3)

SW1 — RESET (ESP_EN, GND)

SW2 — BOOT (IO0, GND)

IMU LSM6DS3TR-C

U8 — LSM6DS3TR-C (3V3, GND, SDO/SA0, SDX, SCX, SDA, SCL, CS, INT1, INT2)

C20 — 100n (3V3, GND)

C21 — 100n (3V3, GND)

R16 — 10k (IMU_SDA, 3V3)

R17 — 10k (IMU_SCL, 3V3)

USB-C NAAR PI 4

J8 — USB-C (VBUS, CC1, CC2, D-, D+, A7, A6, SBU1, SBU2, SHIELD, GND)

J9 — USB-A PI (5V, D-, D+, GND, SHIELD)

C22 — 100n (VBUS, GND)

C23 — 100n (5V, GND)

U9 — USBLC6-2SC6 (ESP_DP, ESP_DM, VBUS)

C24 — 100n (VBUS_ESP, GND)

SENSOR-STROOK (G V S PER GPIO) + I2C

J14 — SENSOR S (GND, 3V3, IO1–IO14)

J15 — SENSOR V (VS, 3V3, IO1–IO14)

J16 — SENSOR G (GND, 3V3, IO1–IO14)

J17 — (GND, 3V3, IMU_SDA, IMU_SCL)

J18 — (GND, 3V3, IMU_SDA, IMU_SCL)

BEVESTIGING

H1 H2 H3 H4

SERVO-BUS A (L1, L2, L3)

U5 — SN74AHCT1G126 (3V3, GND, TX_A, BUSA_DRV)

U6 — SN74LVC1G34 (3V3, GND, BUSA_RXIN, RX_A)

R18 — 120 (BUSA_DRV, BUSA_A)

R19 — 120 (BUSA_A, GND)

R20 — 10k (BUSA_RXIN, 3V3)

R21 — 120 (BUSA_RXIN, BUSA_A)

C25 — 100n (3V3, GND)

C26 — 100n (3V3, GND)

J2 — L1 servo (GND, 12V, BUS_A)

J3 — L2 servo (GND, 12V, BUS_A)

J4 — L3 servo (GND, 12V, BUS_A)

SERVO-BUS B (L4, L5, L6)

U7 — SN74AHCT1G126 (3V3, GND, TX_B, BUSB_DRV)

U10 — SN74LVC1G34 (3V3, GND, BUSB_RXIN, RX_B)

R22 — 120 (BUSB_DRV, BUSB_A)

R23 — 120 (BUSB_A, GND)

R24 — 10k (BUSB_RXIN, 3V3)

R25 — 120 (BUSB_RXIN, BUSB_A)

C27 — 100n (3V3, GND)

C28 — 100n (3V3, GND)

J5 — L4 servo (GND, 12V, BUS_B)

J6 — L5 servo (GND, 12V, BUS_B)

J7 — L6 servo (GND, 12V, BUS_B)

ARACHNE-18 MINI BRAIN v1.0 – ESP32-S3 + MPU-6050 hexapod servo controller for a Raspberry Pi 4 (generated by tools/build_sch.py from tools/design.py)
Servo connectors: 1 = GND, 2 = +12 V, 3 = DATA (Feetech STS half-duplex bus, 1 Mbaud). CHECK the pin order against your cable before plugging in!
Bus A = legs L1 L2 L3 (ESP32 UART1: TX IO17, RX IO18, DE IO16). Bus B = legs L4 L5 L6 (UART2: TX IO10, RX IO11, DE IO9). DE = RTS in RS485 half-duplex mode.
Pi 4 link: USB-C (board) <-> USB-A (Pi) cable on the ESP32-S3 native USB -> /dev/ttyACM0 on the Pi. Pi VBUS only feeds the ESP32 (through D7), never the servos.
5V Pi pads: buck output (5 V, 3 A for the Pi 4) -> Pi GPIO pin 2/4 (+5V) and 6 (GND), or a cut USB-C lead into the Pi power socket.
Buck = TI TPS54560B datasheet design (fig. 32): 400 kHz, 5.0 V 5 A, UVLO start 6.5 V.

ESP32-S3 + MPU-6050, 6 servo-bus legs, USB to Pi 4
Kobe Vanrobaeys – maakleerplek
Sheet: /
File: arachne_mini.kicad_sch
Title: ARACHNE-18 mini brain board
Size: A1 | Date: 2026-10-18 | Rev: 1.0
KiCad E.D.A. 10.0.3 | Id: 1/1